```
DATA Current;
input date CL PR PL;
datalines;
2017.2 25.3 10.3 2.32
;
Run;
```

CODE LOG RESULTS OUTPUT DATA

Table: WORK.CURRENT View: Column names Filter: (none)

Columns

- Select all
- date
- CL
- PR
- PL

| Property | Value |
|---|---|
| Label | CL |
| Name | CL |
| Length | 8 |
| Type | Numeric |
| Format | |
| Informat | |

Total rows: 1 Total columns: 4 Rows 1-1

| | date | CL | PR | PL |
|---|---|---|---|---|
| 1 | 2017.2 | 25.3 | 10.3 | 2.32 |

Messages: 3 User: u64366633

```
Run;
Data Netflow;
input date CL PR PL;
datalines;
2018.1 0.21 0.1 0.05
2018.2 0.22 0.16 0.06
2019.1 .24 0.18 0.08
2019.2 0.28 0.21 0.1
2020.1 0.31 0.23 0.14
;
Run;

Data TransitionMatrix;
```

CODE LOG RESULTS OUTPUT DATA

Table: WORK.NETFLOW View: Column names Filter: (none)

Columns

- Select all
- date
- CL
- PR
- PL

| Property | Value |
|---|---|
| Label | |
| Name | |
| Length | |
| Type | |
| Format | |
| Informat | |

Total rows: 5 Total columns: 4 Rows 1-5

| | date | CL | PR | PL |
|---|---|---|---|---|
| 1 | 2018.1 | 0.21 | 0.1 | 0.05 |
| 2 | 2018.2 | 0.22 | 0.16 | 0.06 |
| 3 | 2019.1 | 0.24 | 0.18 | 0.08 |
| 4 | 2019.2 | 0.28 | 0.21 | 0.1 |
| 5 | 2020.1 | 0.31 | 0.23 | 0.14 |

Messages: 3 User: u64366633

```
Data TransitionMatrix;
input CL PR PL;
datalines;
0.98 0.01 0.01
0.02 0.93 0.05
0.06 0.21 0.73
;
Run;
```

CODE LOG RESULTS OUTPUT DATA

Table: WORK.TRANSITIONMATRIX View: Column names Filter: (none)

Columns

- Select all
- CL
- PR
- PL

| Property | Value |
|---|---|
| Label | |
| Name | |
| Length | |
| Type | |
| Format | |
| Informat | |

Total rows: 3 Total columns: 3 Rows 1-3

| | CL | PR | PL |
|---|---|---|---|
| 1 | 0.98 | 0.01 | 0.01 |
| 2 | 0.02 | 0.93 | 0.05 |
| 3 | 0.06 | 0.21 | 0.73 |

Messages: 3 User: u64366633

```
run;
PROC IML;
use Current; read all into Current;
use Netflow; read all into Netflow;
use TransitionMatrix; read all into TransitionMatrix;


Current = Current [1,2:4];
Netflow = Netflow [,2:4];

Model_2018_1 = Current * TransitionMatrix + Netflow [1,];
Model_2018_2 = Model_2018_1 * TransitionMatrix + Netflow [1,];
Model_2019_1 = Model_2018_2 * TransitionMatrix + Netflow [1,];
Model_2019_2 = Model_2019_1 * TransitionMatrix + Netflow [1,];
Model_2020_1 = Model_2019_2 * TransitionMatrix + Netflow [1,];

Budgetinputs = Model_2018_1//Model_2018_2//Model_2019_1//Model_2019_2//Model_2020_1;

Create Budgetinputs from Budgetinputs;
append from Budgetinputs;
Quit;
```

CODE LOG RESULTS OUTPUT DATA

Table: WORK.BUDGETINPUTS View: Column names Filter: (none)

Columns

- Select all
- COL1
- COL2
- COL3

Total rows: 5 Total columns: 3 Rows 1-5

| | COL1 | COL2 | COL3 |
|---|---|---|---|
| 1 | 25.3492 | 10.4192 | 2.5116 |
| 2 | 25.411296 | 10.570784 | 2.65792 |
| 3 | 25.48396096 | 10.74310528 | 2.77293376 |
| 4 | 25.565519872 | 10.92824361 | 2.8662365184 |
| 5 | 25.654748538 | 11.120831425 | 2.9444200376 |

| Property | Value |
|---|---|
| Label | |
| Name | |
| Length | |
| Type | |
| Format | |
| Informat | |

Messages: 3 User: u64366633

```
Data Output;
Set Budgetinputs (rename=(Col1=Cl Col2=Pr Col3=Pl));
Run;
```

CODE LOG RESULTS OUTPUT DATA

Table: WORK.OUTPUT View: Column names Filter: (none)

Columns

- Select all
- Cl
- Pr
- Pl

| Property | Value |
|---|---|
| Label | |
| Name | |
| Length | |
| Type | |
| Format | |
| Informat | |

Total rows: 5 Total columns: 3 Rows 1-5

| | Cl | Pr | Pl |
|---|---|---|---|
| 1 | 25.3492 | 10.4192 | 2.5116 |
| 2 | 25.411296 | 10.570784 | 2.65792 |
| 3 | 25.48396096 | 10.74310528 | 2.77293376 |
| 4 | 25.565519872 | 10.92824361 | 2.8662365184 |
| 5 | 25.654748538 | 11.120831425 | 2.9444200376 |

Messages: 3 User: u64366633

```
Proc print data=output;
Run;
```

home/u64366633/sasuser.v94/lab5.sas

| Obs | Cl | Pr | Pl |
|---|---|---|---|
| 1 | 25.3492 | 10.4192 | 2.51160 |
| 2 | 25.4113 | 10.5708 | 2.65792 |
| 3 | 25.4840 | 10.7431 | 2.77293 |
| 4 | 25.5655 | 10.9282 | 2.86624 |
| 5 | 25.6547 | 11.1208 | 2.94442 |